How to login to the PC

user: **geomundus001**
password: **ohd9Tiep7Va**

INTELLIGENT
RISK DETECTION
SIDENCA

Intelligent System for DEtection of risks and decision-making support in the event of NAtural CAtastrophes

Parallel Workshop: Wildfire Risk Mapping and Fire Spread Modelling with Open Earth Observation Data

Pablo Juan Verdoy

GeoMundus Conference 17 October 2026

GEOMUNDUS Conference — 18th Edition —

GENERALITAT VALENCIANA | IVACE+i | Finançat per la Unió Europea | Fons Europeus

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INTELLIGENT RISK DETECTION
SIDENCA

SIDENCA WHY?

The Valencian region faces interconnected climate hazards

FLOODS

DANA, Oct 2024

More than 200 deaths and a poorly coordinated response between agencies.

WILDFIRES

>350,000 ha

burned in large fires, 1986–2019; recent cases include Bejís and Vall d'Ebo.

HEATWAVES

44.4 °C

in Carcaixent (2015); August 2021 set a record since 1950.

ALL HAZARDS AT A PLACE

Heat feeds drought, drought feeds fire — yet warning and management systems remain **fragmented, reactive and poorly interoperable**.

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INTELLIGENT RISK DETECTION
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THE PROJECT

SIDENCA: multi-Hazard decision support for the Valencian region

An interoperable, scalable platform combining EO, drones, IoT sensors, predictive models and AI for **wildfires, floods, droughts and heatwaves** — designed with emergency coordination centres and civil protection.

Objective	What it delivers
Modelling	Event propagation, evolution and severity (INLA-SPDE)
EO processing	Automated products from satellite and drone data
Integration	Semantic interoperability, near-real-time data
Co-design	Requirements defined with emergency services
Validation	Testing in real emergency scenarios
Edge analytics	Alerts under limited connectivity

Consortium · 2025–2027

Albavalor (coordinator) · AINIA · Innovageo · Laberit · Universitat Jaume I · Universitat de València

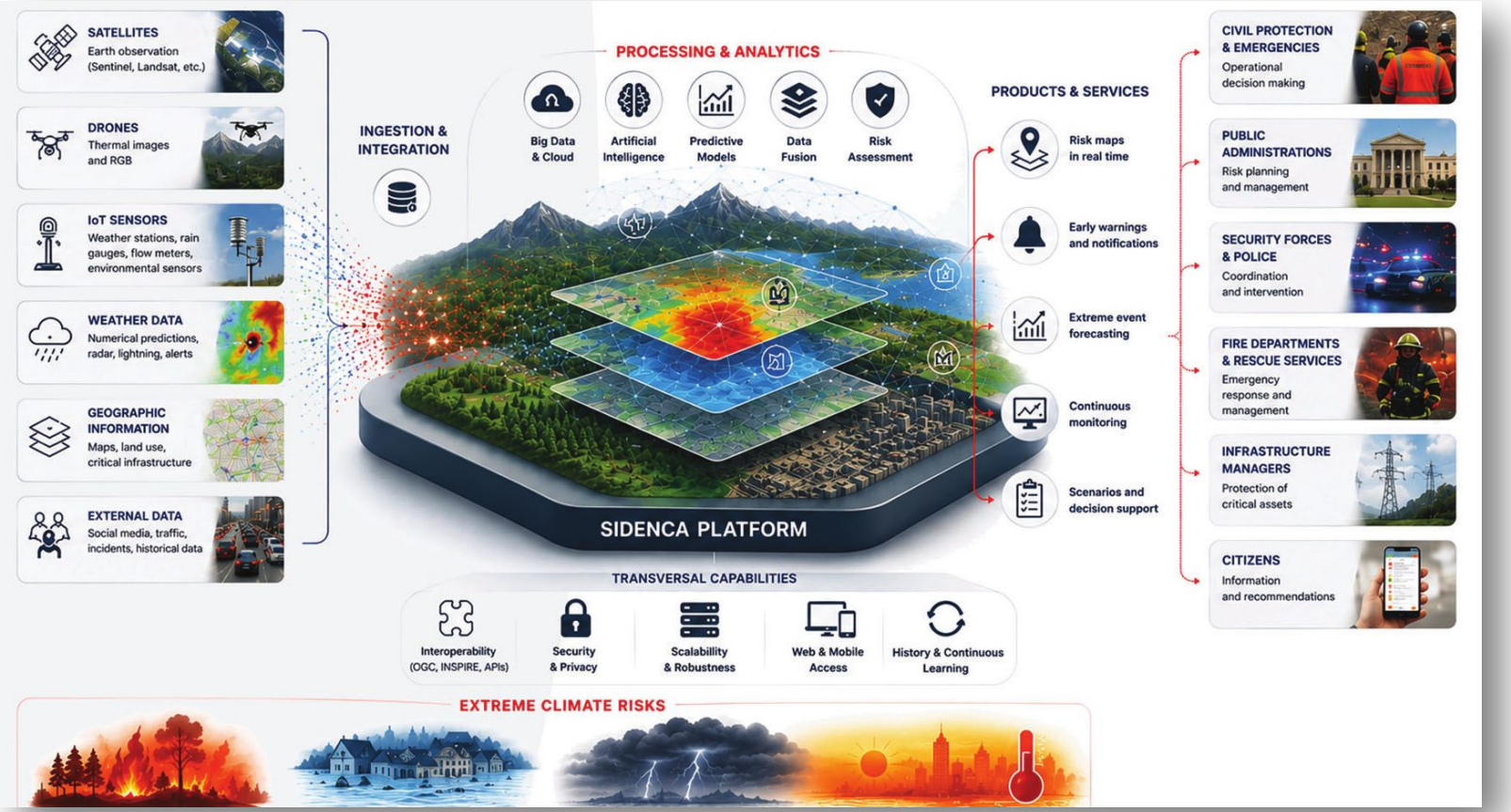
THIS TALK

The EO wildfire workflow — data acquisition and processing that feeds the spatial models developed at UJI.

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SIDENCA OPERATIONAL MODEL

One platform, from data sources to emergency end-users



In

Satellites, drones, IoT sensors, weather, GIS, citizen data

Inside

Semantic data integration, INLA-SPDE and AI models, edge nodes

Out

Risk maps, early warnings, monitoring, scenarios — web and mobile geoportal

For

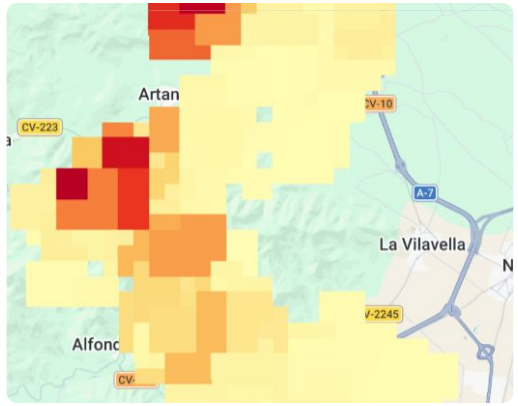
112, fire services, civil protection, municipalities, citizens

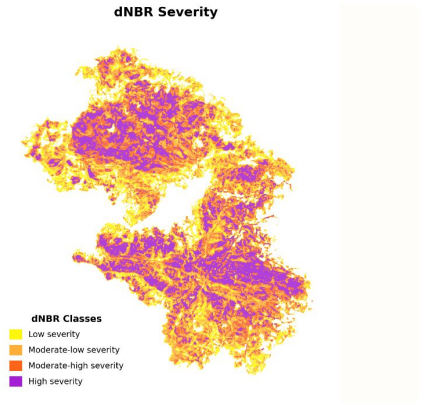
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INTELLIGENT RISK DETECTION
SIDENCA

SIDENCA DEMONSTRATORS, JULY 2026

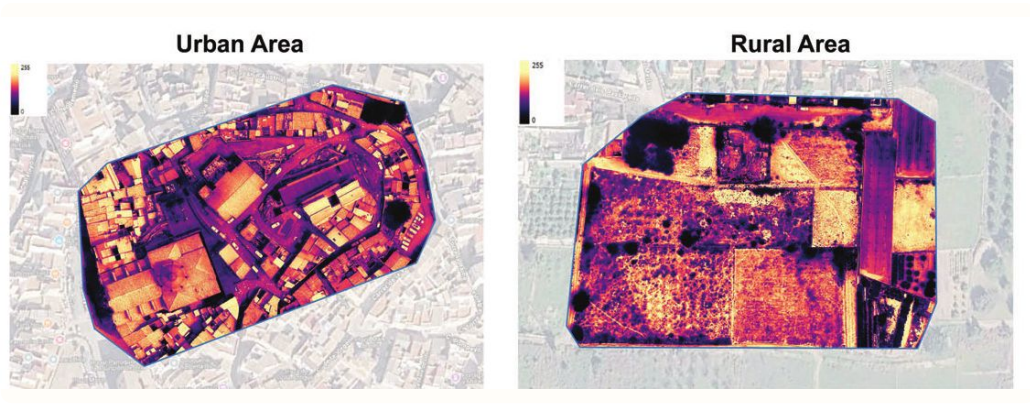
First demonstrators ran on real events this summer





Wildfire · La Vall d'Uixó

Satellites, from active-fire intensity to burn severity — the rest of this talk.



Heatwave · Llíria

Thermal drone flights (22 July): urban vs. rural hotspots for heat-exposure assessment.

Prevent	Prepare	Respond	Recover	Decide
Risk baselines	Scenarios, resources	Detection, alerts	Impact, recovery	Shared picture

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INTELLIGENT RISK DETECTION
SIDENCA

FOCUS: WILDFIRES

During a wildfire, managers need one coherent picture

The setting

Hot, dry summers, rugged terrain and a wide wildland–urban interface drive fast spread and high exposure across the Mediterranean.

The gap

EFFIS, CEMS and FIRMS deliver excellent but separate products, each for one stage of the fire cycle — not an incident-specific chain.

The constraints

Revisit time, clouds and smoke, overpass timing, and 250–500 m burned-area products that miss fragmented Mediterranean fires.

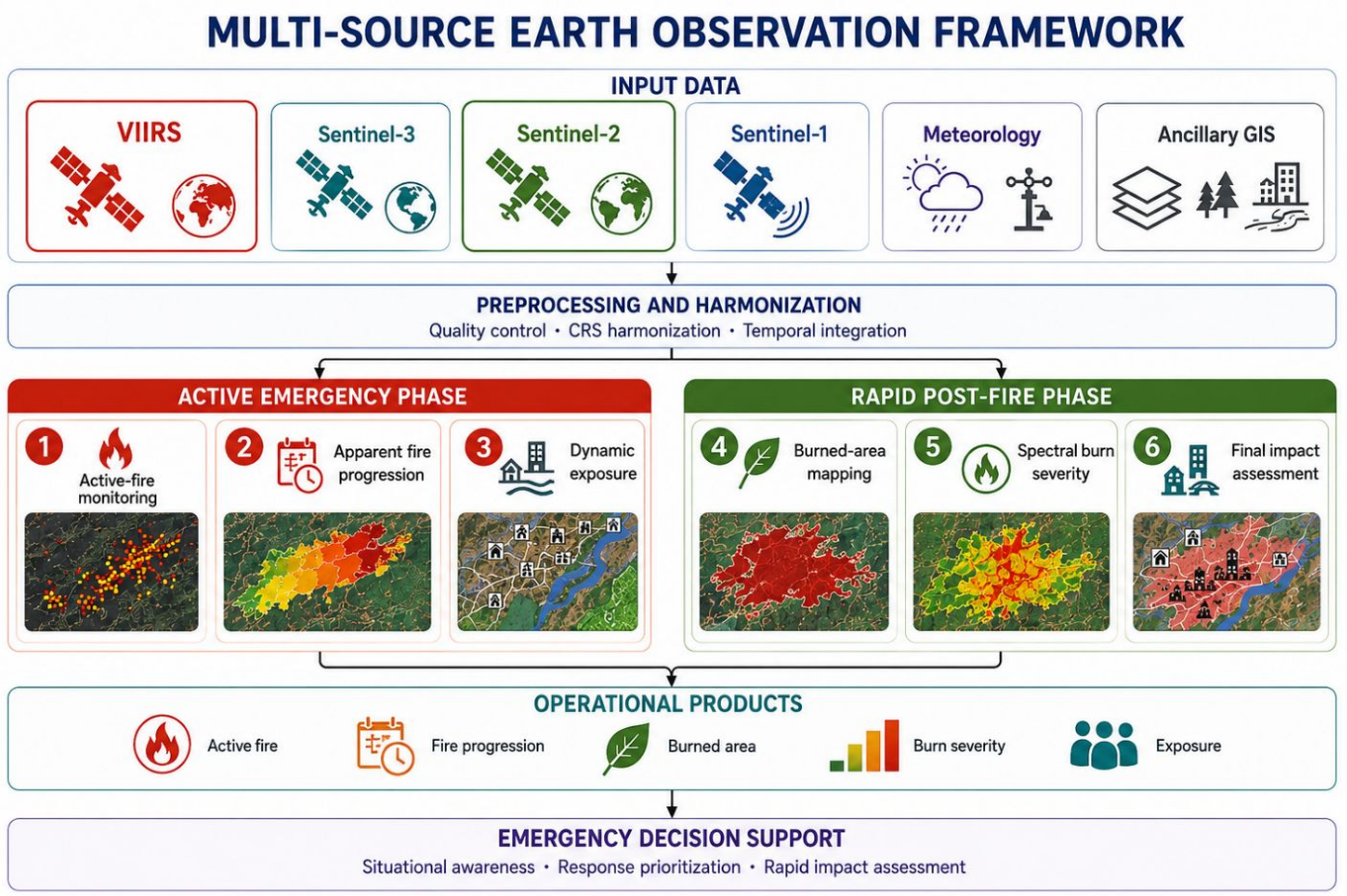
OUR QUESTION

Can heterogeneous open EO data be integrated into a **reproducible workflow** that supports near-real-time wildfire emergency management — from ignition to post-fire impact?

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THE FRAMEWORK

One workflow, two phases, six operational products



ACTIVE EMERGENCY

1 · Active-fire monitoring

2 · Apparent fire progression

3 · Dynamic exposure

RAPID POST-FIRE

4 · Burned-area mapping

5 · Spectral burn severity

6 · Final impact assessment

Not a new satellite product: an integration chain that is reproducible and transferable to other regions.

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INTELLIGENT RISK DETECTION
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OPEN DATA STACKS

Everything runs on open, freely available data

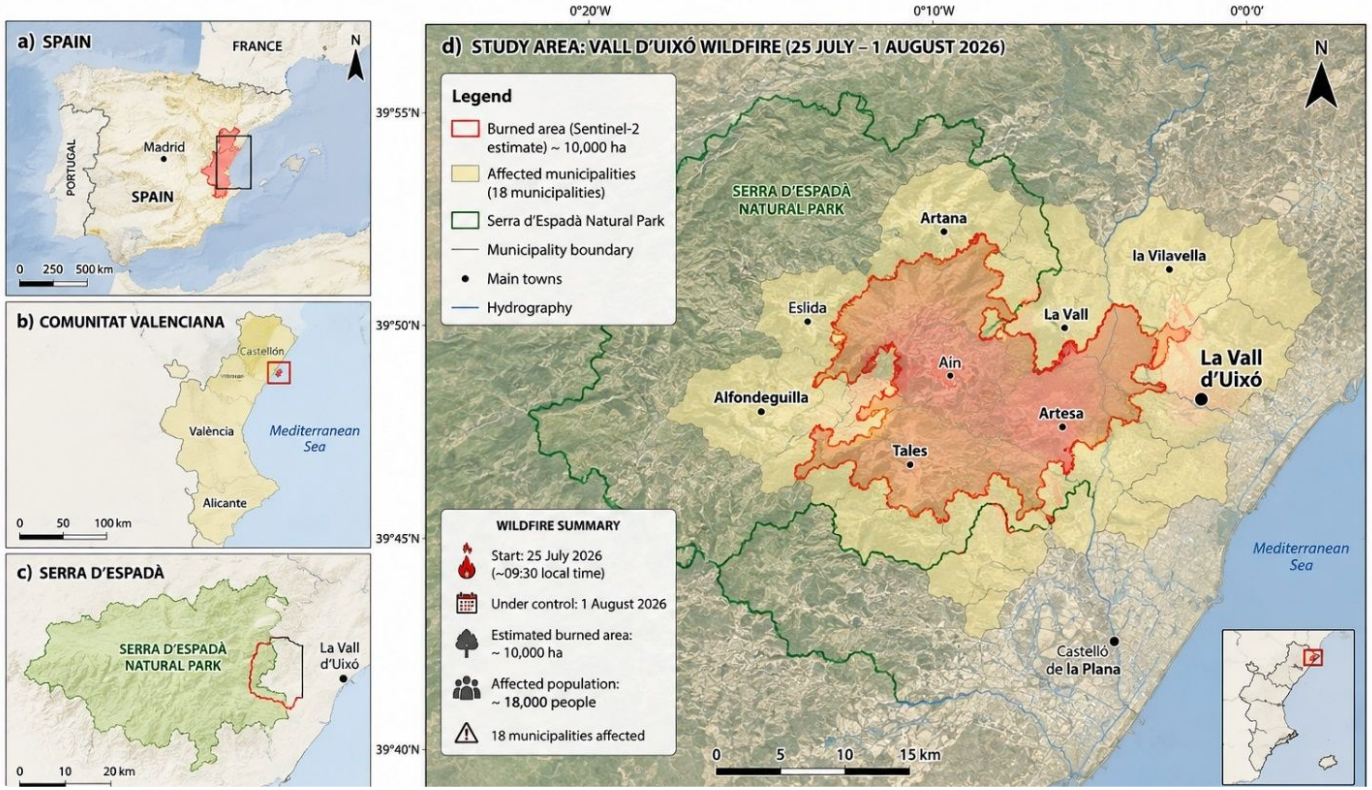
Component	Dataset	Resolution	Role in the workflow
Active fire	VIIRS 375 m (Suomi-NPP + NOAA-20)	375 m · NRT	Detection, chronology, FRP, persistence
Active fire	Sentinel-3 SLSTR FRP	~1 km · daily	Complementary thermal activity
Burned area	Sentinel-2 MSI L2A	10–20 m · ~5 days	Perimeter, dNBR / RBR severity
Fire weather	ERA5-Land (17 variables)	~9 km · hourly	FWI, drivers (≈5-day latency)
Fire weather	AEMET / AVAMET stations	Point · hourly	Real-time conditions
Fuels	GVA fuel model map (Scott & Burgan)	10 m	Fuel type and continuity
Terrain	ICV LiDAR DTM	10 m	Slope, aspect, TRI, TPI
History	GVA fire perimeters 1993–2024	Vector	Fire recurrence
Exposure	WorldPop · IGN transport · Natura 2000	100 m · vector	People, roads, protected areas
Response	Water points · tanks · fuel breaks (GVA)	Vector	Suppression resources nearby

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INTELLIGENT RISK DETECTION
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CASE STUDY

Vall d'Uixó /Serrà d'Espadà wildfire, 25 July- 1 August



~10,000 ha
burned (Sentinel-2 estimate)

~18,000
residents evacuated or confined

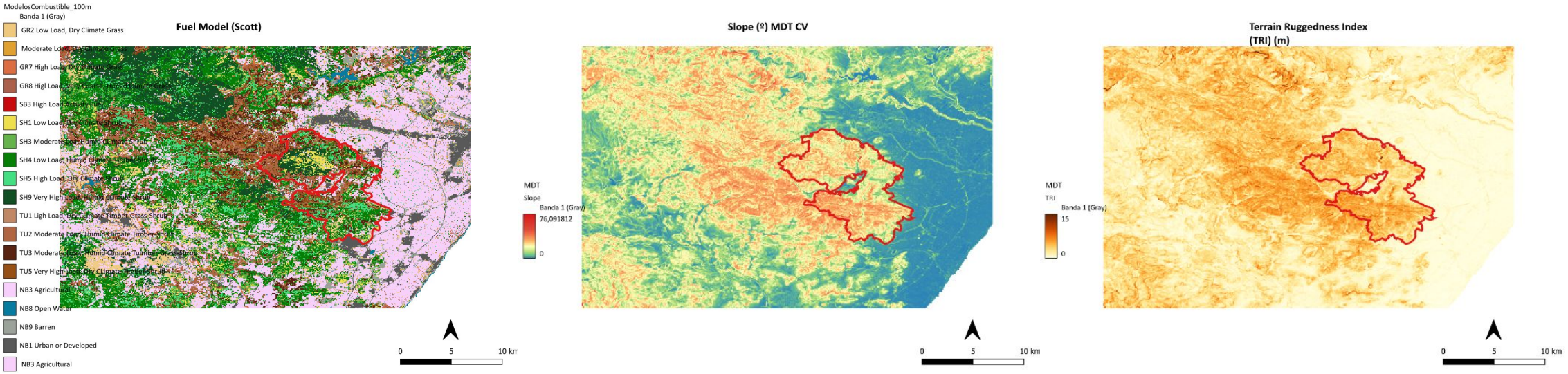
18
municipalities affected

Pine forests and shrublands, 100–1,000 m relief, strong westerly winds. UME deployed; Copernicus EMS activated (EMSR905).

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BEFORE THE FIRE- FUELS AND TERRAIN

Continuous fuels on rugged terrain set the stage



Fuel models · 10 m

14 burnable Scott & Burgan classes; timber-understory and shrub fuels dominate inside the perimeter.

Slope · LiDAR DTM

Steep slopes tilt flames toward unburned fuel and speed up uphill spread.

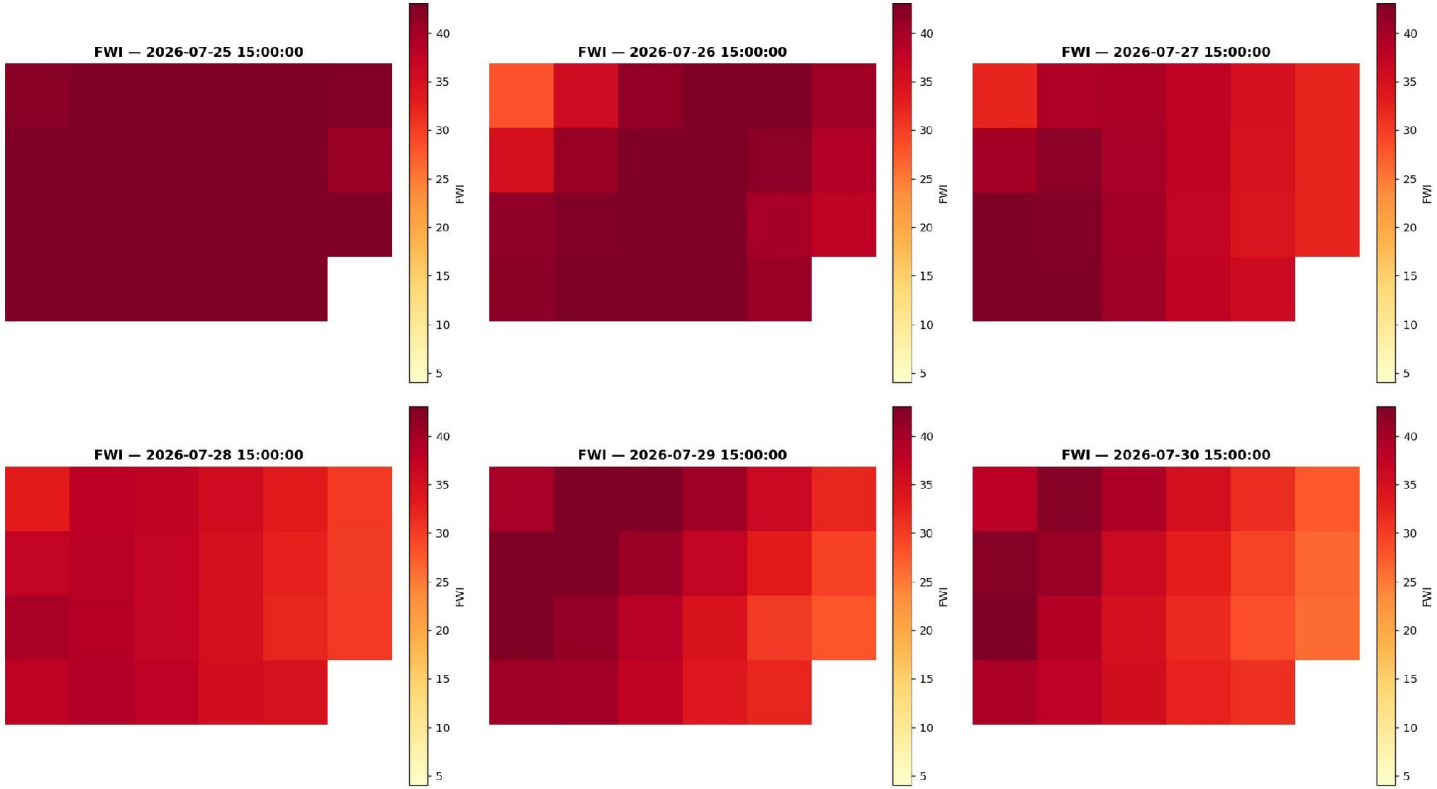
Terrain ruggedness

The fire sits in an intermediate-to-high TRI zone: hard ground access and wind channelling.

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BEFORE THE FIRE- FIRE WEATHER

Continuous fuels on rugged terrain set the stage



Daily FWI at 15:00 local, 25–30 July 2026 · ERA5-Land grid (~9 km)

FWI class	Range
Low	5.2 – 11.2
Moderate	11.2 – 21.3
High	21.3 – 38.0
Very high	38.0 – 50.0
Extreme	≥ 50

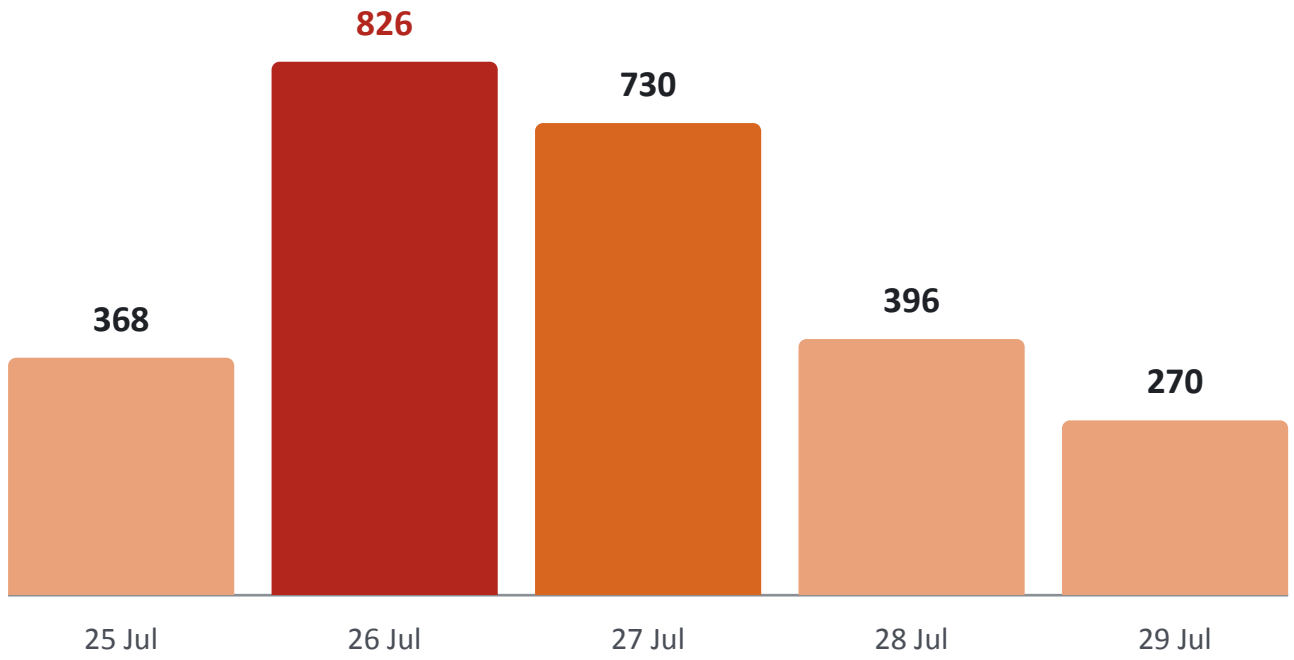
Canadian FWI computed per pixel from ERA5-Land temperature, humidity, wind, and 24-h precipitation; most cells fall into the high-to-very-high classes.

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DURING THE FIRE – VIIRS 375 m

Fire activity peaked on day two, then declined

Reliable detections per day · Suomi-NPP + NOAA-20 merged



2,590
fire pixels

352 MW
max FRP

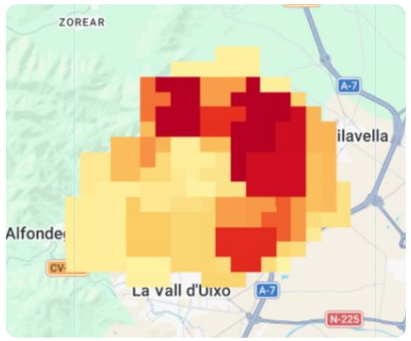
Overpass	Pixels	Mean FRP	Max FRP
Day	1,343	36.6 MW	351.7 MW
Night	1,247	7.1 MW	63.7 MW

Day FRP is ~5× night FRP — partly overpass timing, so not direct evidence of diurnal behaviour or suppression effect.

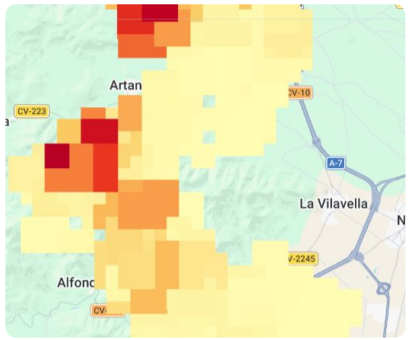
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DURING THE FIRE – SPREAD CHRONOLOGY

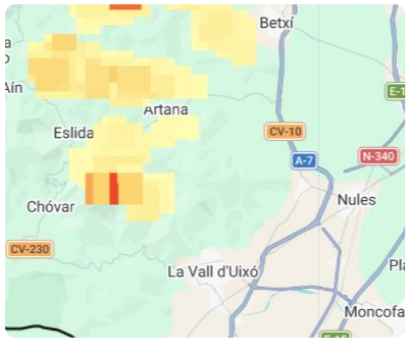
Daily VIIRS maps reconstruct how the fire spread



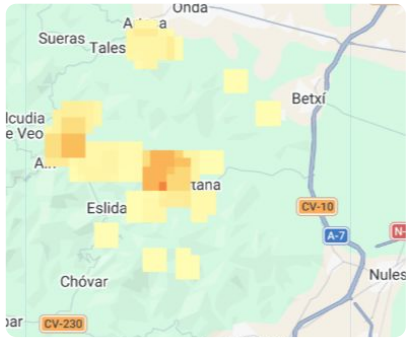
25 Jul · ignition



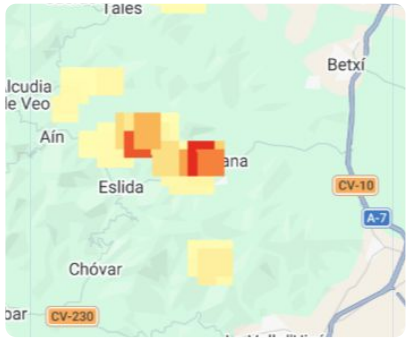
26 Jul · peak



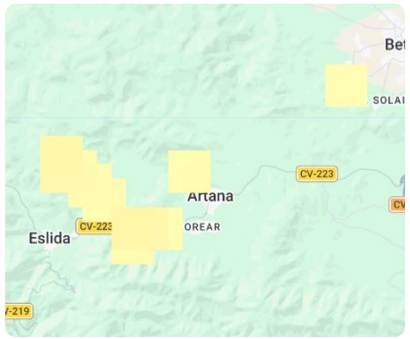
27 Jul



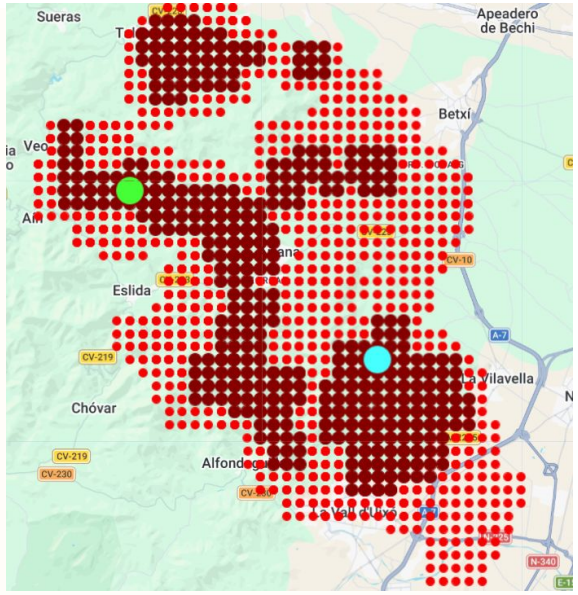
28 Jul



29 Jul



30 Jul · residual

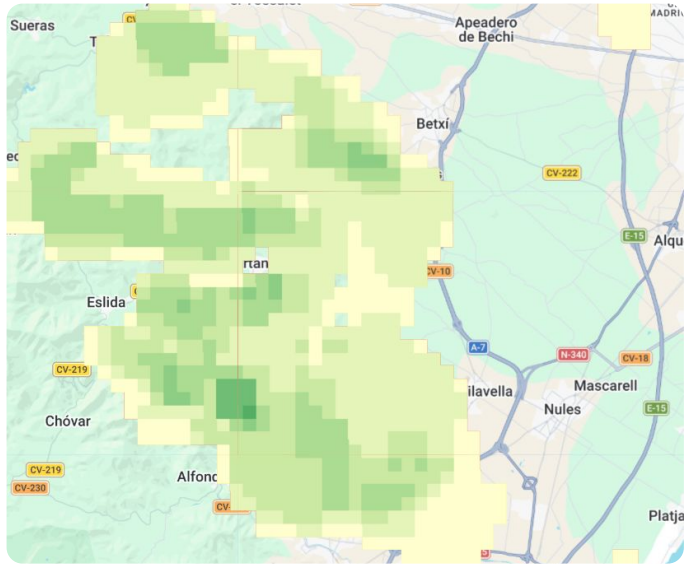


All fire pixels (375 m): first reliable detection in **blue**, last in **green** — an apparent spread from the south-east toward the north-west.

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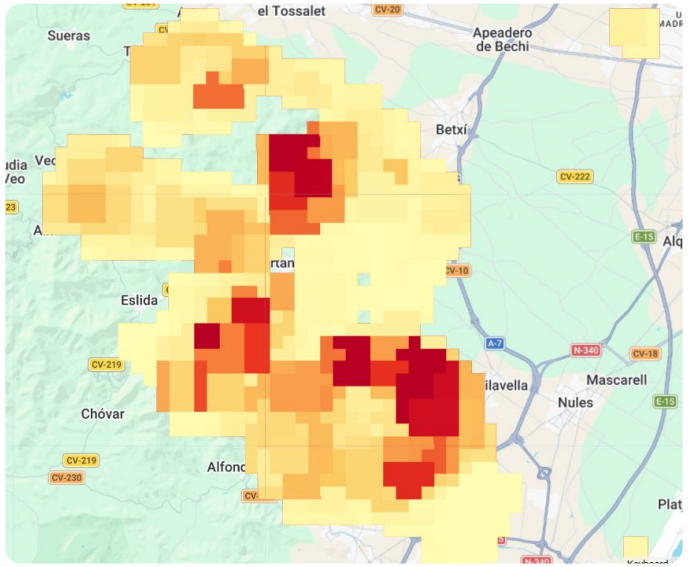
DURING THE FIRE – INTENSITY AND PERSISTENCE

Three composites show where, how hard and how long it burned



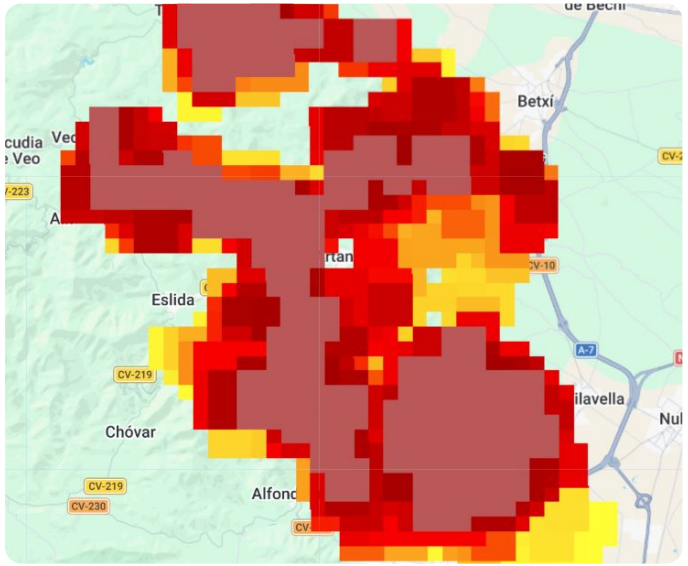
Detection density → persistence

Overpasses in which each pixel was active. A proxy, not fire duration.



Maximum FRP → intensity

High-energy clusters in the central and southern sectors; lower values at the edges.



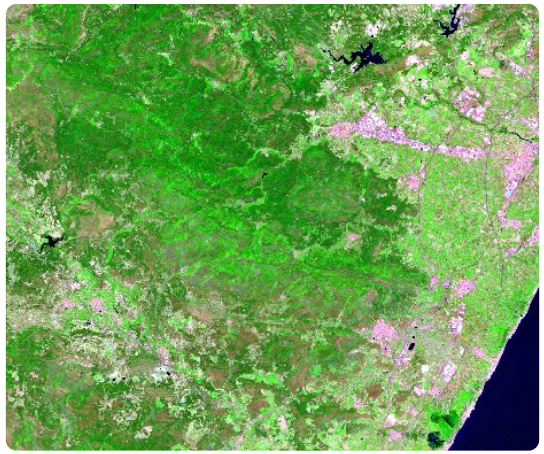
Max I4 brightness → thermal state

Peak thermal emission per pixel; mean 341 K, maximum 367 K.

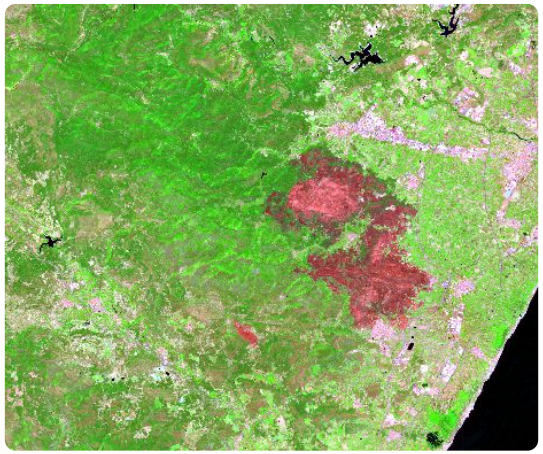
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AFTER THE FIRE – BURNED AREA

Sentinel-2 dNBR and RBR match the official perimeter best



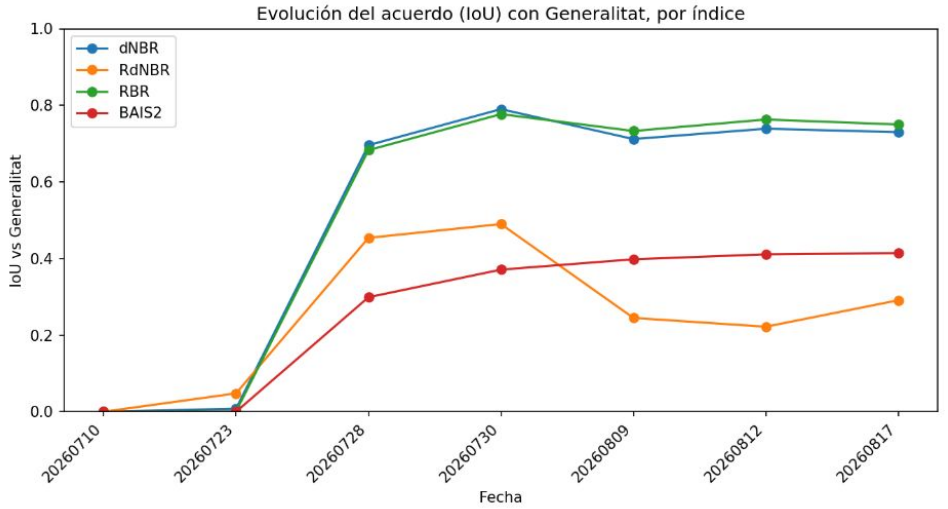
Pre-fire



Post-fire

≈0.75
IoU, dNBR and RBR

<0.5
IoU, RdNBR and BAIS2

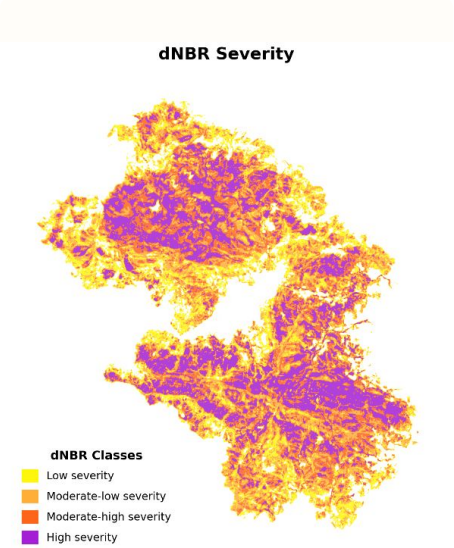


Agreement (IoU) with the Generalitat Valenciana perimeter for each Sentinel-2 date. It stabilises once the smoke clears, from 30 July.

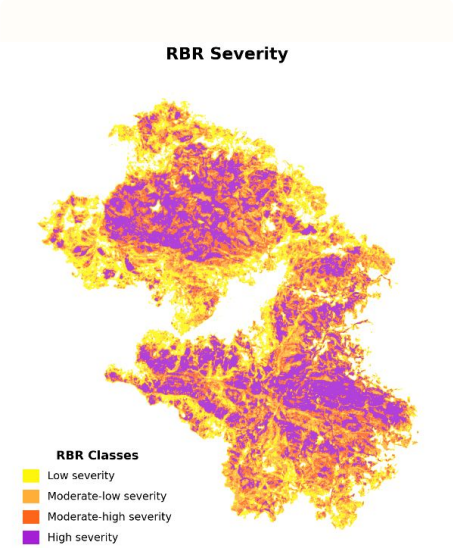
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AFTER THE FIRE – BURN SEVERITY

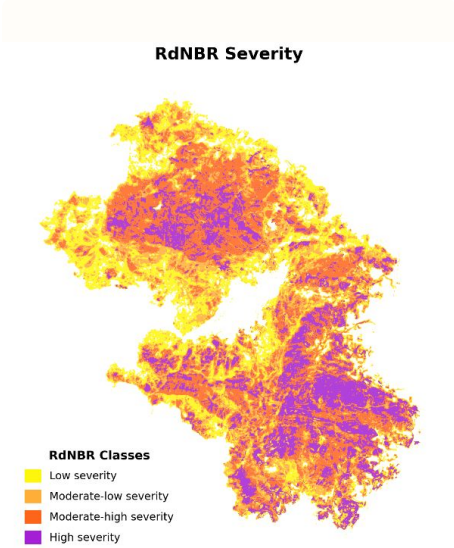
A third or more of the burned áreas shows high severity



dNBR



RBR



RdNBR

Index	Burned (ha)	High
dNBR	8,681	33.9 %
RBR	8,870	36.9 %
RdNBR	9,154	40.9 %

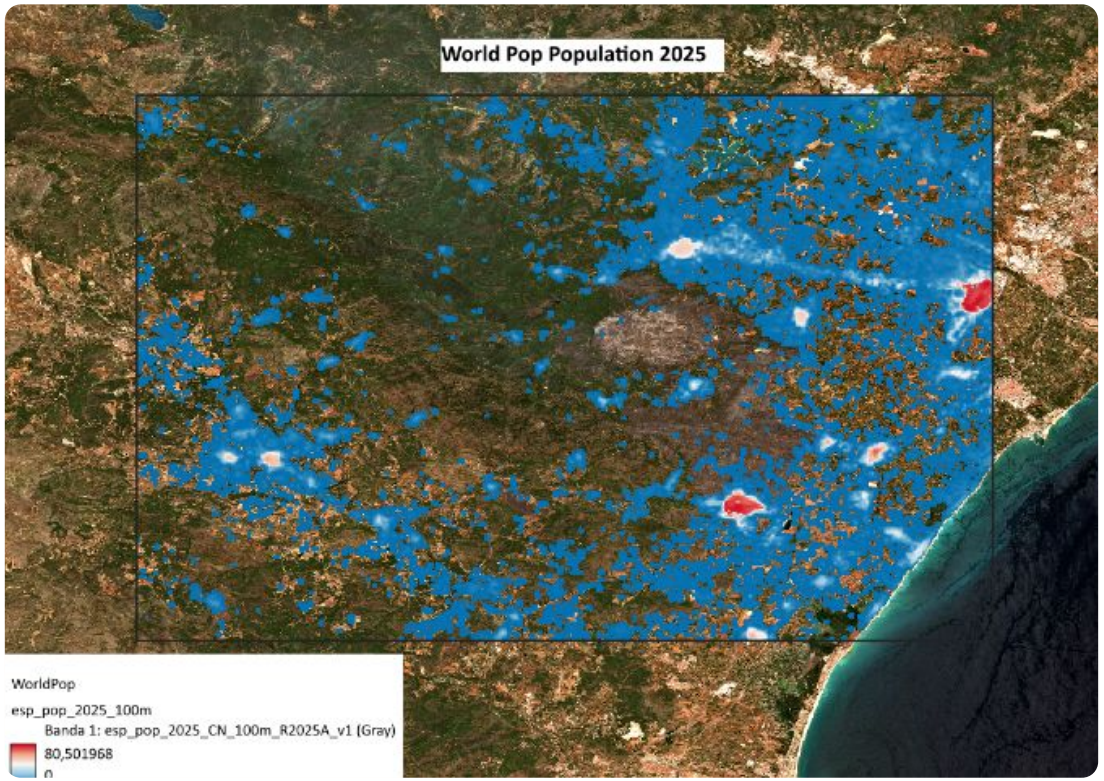
Four classes: low, moderate-low, moderate-high, high. The spatial pattern is consistent; RdNBR pushes more area into the high class.

Severity is **spectral change**, not yet field-validated (CBI).

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DURING AND AFTER – EXPOSURE AND RESPONSE

The fire extent is turned into decisio-ready indicators



WorldPop 2025 population, 100 m

People
Residents inside the fire extent

Roads
Km affected by class (IGN)

Nature
Natura 2000 and park area, ha and %

Water
Points and tanks within 1, 2, 5 km; nearest distance

Fuel breaks
Area intersected vs. observed spread

Trails
Km of XINM routes affected

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INTELLIGENT RISK DETECTION
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POSITIONING

SIDENCA complements existing services rather tan replacing them

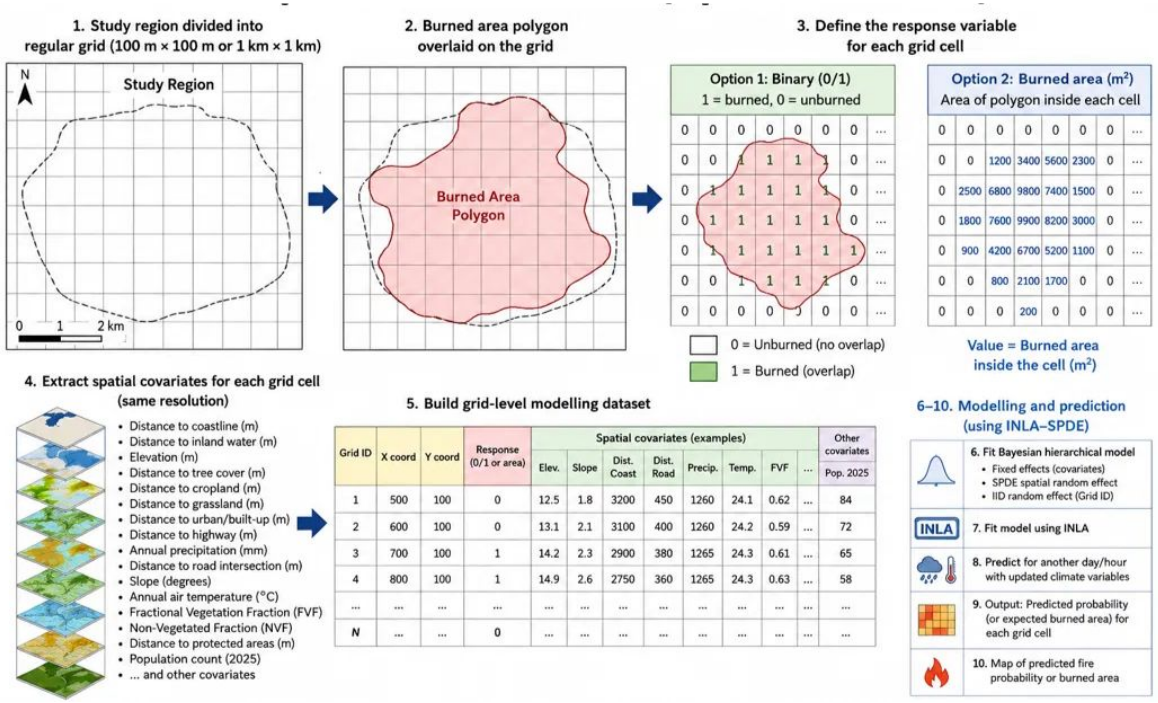
Capability	FIRMS	EFFIS	CEMS	GVA reports	SIDENCA
NRT active-fire detection	Yes	Yes	Partial	No	Yes
Fire Radiative Power	Yes	Partial	No	No	Yes
Fire chronology	No	No	No	Manual	Yes
Burn severity	No	Partial	Partial	Yes	Yes
Exposure analysis	No	Partial	Yes	Partial	Yes
Fuel model integration	No	No	No	Yes	Yes
Automated workflow	Yes	Partial	Partial	No	Yes
Detection → impact chain	No	No	No	No	Yes

FIRMS = NASA hotspots · EFFIS = European Forest Fire Information System · CEMS = Copernicus EMS Rapid Mapping · GVA = Generalitat Valenciana post-fire reports

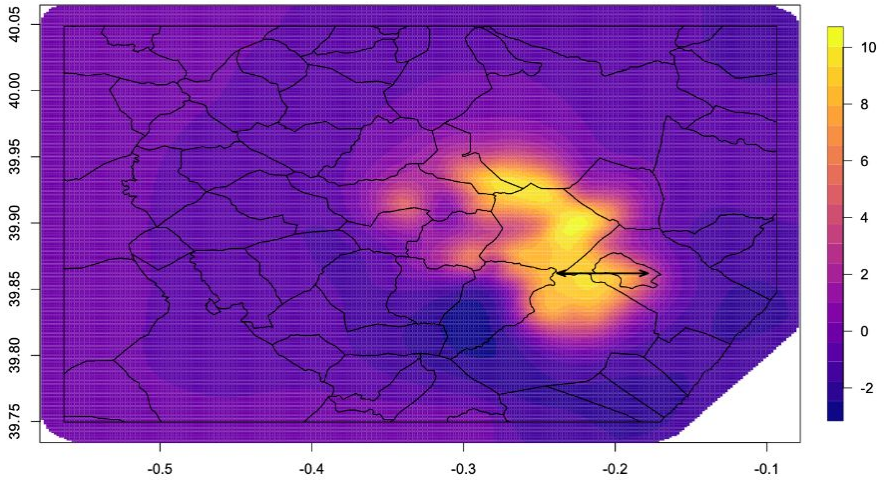
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TOWARDS RISK MAPPING – WORK IN PROGRESS

EO products become covariates for SPDE-INLA risk models



Grid → burned response → covariates → INLA-SPDE → probability of burning



Spatial random effect: what terrain, fuel and human-access covariates do not explain — with explicit uncertainty.

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LESSONS LEARNED

What worked, what limits us, and what comes next

What worked

- Merged VIIRS gives a usable chronology within hours
- dNBR and RBR agree with the official perimeter (IoU ≈ 0.75)
- Fully open data: reproducible and transferable

Limits

- 375 m pixels, few overpasses: apparent spread only
- Smoke and clouds delay Sentinel-2
- ERA5-Land: 9 km and ~5-day latency
- Severity not yet field-validated

Next

- Sentinel-1 SAR and Sentinel-3 SLSTR to see through smoke
- Feed SPDE-INLA risk and spread models
- Deploy in the SIDENCA geoportal; test on more fires

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INTELLIGENT RISK DETECTION
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TAKE-HOME MESSAGE

Open EO data can already follow a wildfire from danger to impact. The valu is in the chain

Fuels and fire weather → VIIRS chronology and intensity → Sentinel-2 burned area and severity → exposure → risk models

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deja
dudas
compartir
ideas
recetas
preguntas
un
comentario
comentar

Meteorological and pollution data

Wildfire Risk Mapping and Fire Spread Modelling with Open Earth Observation Data

Sergi Trilles
16 October 2026

Introduction

- **Weather:** temperature, humidity, rainfall and wind.
- **Air quality:** concentrations of particles and gases in the air.
- **Collection:** on-site sensors at monitoring stations and sensor networks.
- **Outputs:** measurements linked to time and location, organised into time series and maps.

- **GEOTEC applications:** forecasting, anomaly detection and environmental monitoring.



AEMET's observation network

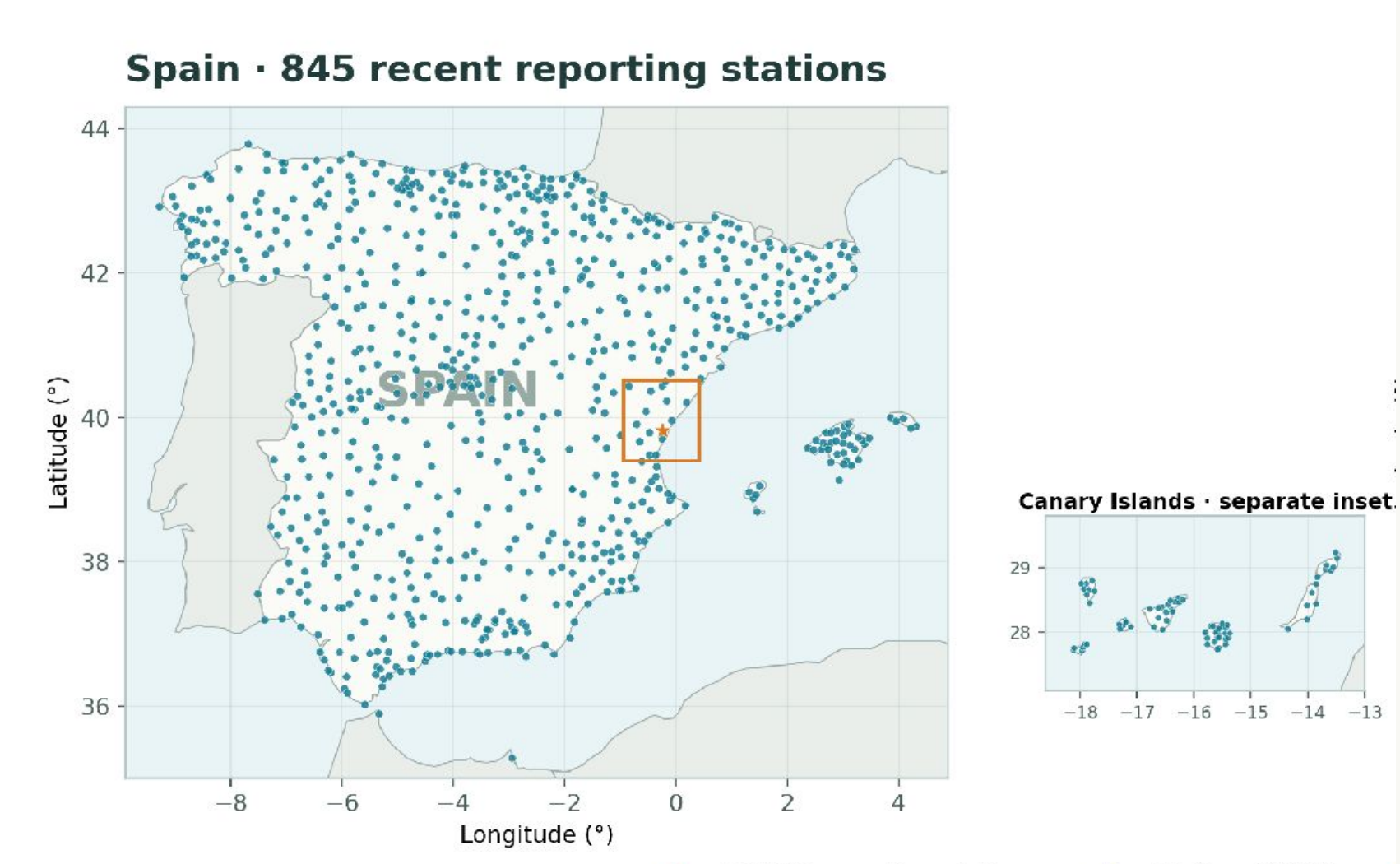
- **National coverage**: in-situ meteorological observations across Spain
- Meteorological stations: **849 stations** and 75 observing sites with AEMET staff
- Collaborator network: 1,974 collaborator-supported stations, mainly measuring rainfall and temperature
- Complementary systems: **weather radar, satellites, lightning detectors and radiosondes**

Automatic weather station



1. **Weather shelter**: protects temperature and humidity sensors
2. **Wind mast**: supports wind speed and direction sensors
3. **Equipment cabinet**: records and transmits measurements
4. **Rain gauge**: measures rainfall

AEMET station locations



Examples of AEMET stations in Castellón



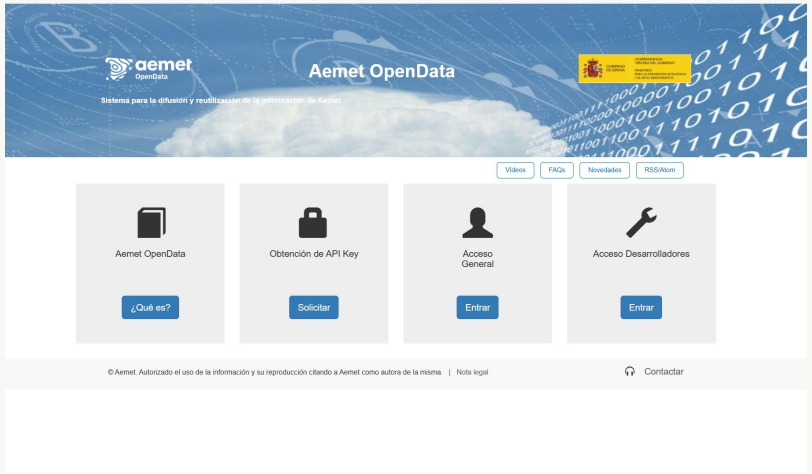
Atzeneta del Maestrat station



Castellón observatory

AEMET meteorological data

- Variables: **temperature, humidity, rainfall, wind and pressure**.
- Records: **recent hourly observations and historical daily records**.
- Availability: depends on the station, variable and period.
- Access: website consultation and OpenData downloads using an API key.



https://opendata.aemet.es

AEMET products

Product	Access
Historical daily data	Daily records from past years
Recent observations API	Hourly observations from the last 12 hours
Observation website	Last 24 hours and daily summaries for 7 days
Forecasts and warnings	Expected weather conditions and alerts

AEMET hourly observations (API)

Variable	What the hourly value means
Temperature / humidity	Measurement at the observation time
Mean wind	Average over the previous 10 minutes
Maximum wind gust	Highest gust during the previous hour
Rainfall	Total accumulated during the previous hour

An hourly value can represent an instant, an average, a maximum or an accumulation.

Air-quality data sources

Source	Data	Access route
European Environment Agency	Provides air-quality measurements reported by European countries, supporting comparisons between locations.	https://www.eea.europa.eu/en/datahub/datahubitem-view/778ef9f5-6293-4846-badd-56a29c70880d
MITECO - Spain's environment ministry	Collects official air-quality observations from regional and local monitoring networks across Spain.	https://www.miteco.gob.es/es/calidad-y-evaluacion-ambiental/temas/atmosfera-y-calidad-del-aire/evaluacion-y-datos-de-calidad-del-aire/datos.html
GVA regional network	Regional monitoring network. Hourly and daily pollutant measurements from stations across the Valencian Community	https://rvvcca.pica.gva.es/es/datos-de-la-rvvcca
Madrid air quality network	Municipal monitoring network. Hourly pollutant measurements for Madrid city, with historical records since 2001.	https://datos.madrid.es/dataset/201200-0-calidad-aire-horario

Weather stations and satellites

Source	Advantages	Limitations
In-situ stations	Direct local measurements; frequent observations	Uneven spatial coverage; missing records
Remote sensing	Broad coverage; captures spatial patterns	Indirect estimates; revisit intervals; cloud effects for some sensors

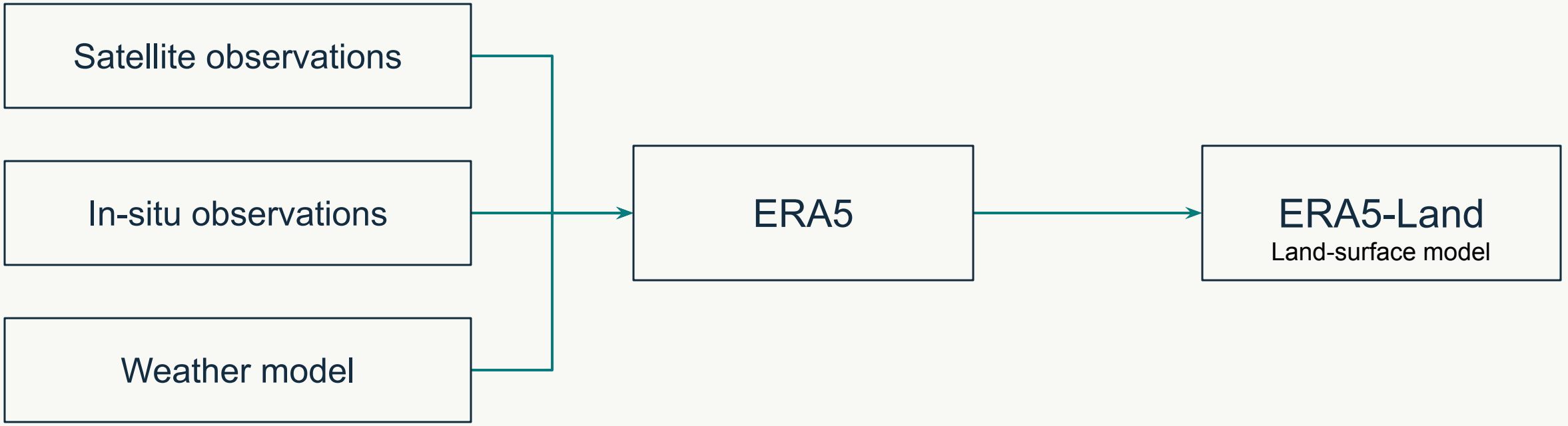
Why use ERA5?

ERA5 combines observations with a weather model to reconstruct past weather, providing:

- **Complete spatial coverage**: estimates across the study area, including places without weather stations.
- **Historical and recent data**: hourly estimates from 1940 to approximately five days ago.
- **A regular grid**: weather variables organised consistently across space and time.

We can compare ERA5 estimates with local AEMET measurements to assess how well they represent local conditions.

From observations to ERA5 / ERA5-Land



Observations influence ERA5-Land indirectly, through the weather information provided by ERA5.

Labels: the official fire register

Item	Details
Dataset	Incendios forestales de la Comunitat Valenciana, 1993 to 2024
Published by	Institut Cartogràfic Valencià, with the fire service
One record	One fire: the area it burnt, plus its official report
Mapping	Satellite images, GPS surveys and aerial photos, at 1:5,000
Fields we use	Detection date and time, reported start point, cause
Format	Polygons in ETRS89 UTM 30N (EPSG:25830), licence CC BY 4.0

Source: dadesobertes.gva.es. The official record, not a satellite hotspot product.

Every input at a glance

Variable	Source	Native resolution	How often it changes	What each sample gets
Weather, 5 values	ERA5-Land	0.1°, about 8.5 × 11 km	Daily, built from hourly data	7 days, ending on the sample day
Elevation	SRTM	About 30 m	Fixed, measured in 2000	A 32 × 32 patch of 100 m cells
Slope, eastness, northness	Computed from elevation	100 m	Fixed	A 32 × 32 patch
Land cover, 8 classes	CORINE	100 m, nothing under 25 ha	Every 6 years, 2000 to 2018	A patch from the edition before the sample year

Five weather values, every day

Variable	Unit	What it is
Maximum temperature	°C	The hottest hour of the day, 2 m above the ground
Mean temperature	°C	The average of the day's 24 hourly values
Relative humidity	%	How dry the air is, worked out from temperature and dewpoint
Wind speed	m/s	Daily mean at 10 m, from the east and north wind components
Rain	mm	The day's total. Most days have none, so the model sees it on a log scale

Each sample gets these five values for the 7 days ending on its date: a table of 7 rows by 5 columns.

Terrain and land cover

Terrain

SRTM, fixed

- **Elevation:** height above sea level
- **Slope:** steepness, in degrees
- **Eastness, northness:** which way the slope faces

Land cover

CORINE, every 6 years

- **8 groups:** built-up, farmland, forest, shrub and grass, sparse or burnt, wetland, water, no data
- **Edition:** the one before the sample year, so it never sees the fire

All three are cut as a 32 × 32 patch of 100 m cells (3.2 km) around each sample.

Each input has its own clock

Example: the Bejís fire, detected on 15 August 2022. This is what the model sees for it.

Input	Changes	Version used for this fire
Weather	Every day	9 to 15 August 2022, one row per day
Land cover	Every 6 years	CORINE 2018 edition
Terrain	Never	The same layers for every sample

The rule: an input may only use information that existed before the sample date.

One 100 m grid, four native scales

Input and native size	Versus one 100 m cell	How it reaches the grid	What is lost
SRTM elevation, 30 m	About 11 pixels per cell	Averaged	Detail under 100 m
CORINE land cover, 100 m	One pixel per cell	Copied, nearest class	Patches under 25 ha
ERA5-Land weather, 9 km	About 9,500 cells per pixel	Nearest cell, for now	Local weather differences

A model output at 100 m is only as detailed as the inputs that vary at 100 m.